

day 27 | 251031 F

herein

We are now ready to handle 2nd order differential equations. The method will be similar to coupled first order equations because that is exactly how we will treat any second order equation: as a set of coupled first order equations like below.

Let's take an example second order equation and look at the technique. We will take a mass on a spring under damped conditions:

$$ma = -cv - kx$$

The first thing we want to do is put this in terms of all of the derivative:

$$m \frac{d^2 x}{dt^2} = -c \frac{dx}{dt} - kx$$

Then we want to isolate the second derivative.

$$\frac{d^2 x}{dt^2} = -\frac{c}{m} \frac{dx}{dt} - \frac{k}{m} x$$

Then we want to break this up into two first order differential equations. And this looks a little bit backwards, but we need to realize that we just replaced v with $\frac{dx}{dt}$ so what if we went backwards that in terms of the substitution of variables.

$$\frac{dv}{dt} = -\frac{c}{m} v - \frac{k}{m} x$$

and then

$$\frac{dx}{dt} = v$$

So now we can run this exactly the same way that we did coupled first order equations, because that is what they are! And we can choose almost any kind of differential equation that we want to. To start off with we may just choose an object falling from some height. The initial height is an important initial value and the fact that we are dropping it is important since the initial velocity will be equal to zero.

$$a = -g \Rightarrow \frac{d^2 x}{dt^2} \Rightarrow \frac{dv}{dt} = -g \quad \text{and} \quad \frac{dx}{dt} = v$$

And this same logic can be applied to other sets, so we could cover linear drag:

$$\frac{dv}{dt} = -g - \frac{c}{m}v \quad \text{and} \quad \frac{dx}{dt} = v$$

Or we can do quadratic drag but you do have to be more careful since the signs are more tricky:

$$\frac{dv}{dt} = -g + \frac{c}{m}v^2 \quad \text{and} \quad \frac{dx}{dt} = v$$

For this case you have to watch out for the *SIGNS* of the velocity because they are squared and there is not an easy way to handle them if the object comes to rest and changes direction.

Work on exercise 8.5 after you have reviewed the example 8.6 and exercise 8.4.

```
In [11]: %matplotlib ipynpl
import numpy as np
import matplotlib.pyplot as plt
```

```
In [22]: # define the differential equation
def f(r, t): # don't forget r is a vector!
    x = r[0]
    v = r[1]
    g = 9.8 # N/kg
    fx = v # this is the dx/dt equation!
    fv = -g # this is the dy/dt equation!
    return(np.array([fx, fv]))

# define the boundary condition
a = 0.0
b = 1.414
N = 100
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([10.0, 0.0]) # initial condition

tpoints = np.arange(a, b, dt)

xpoints = []
vpoints = []

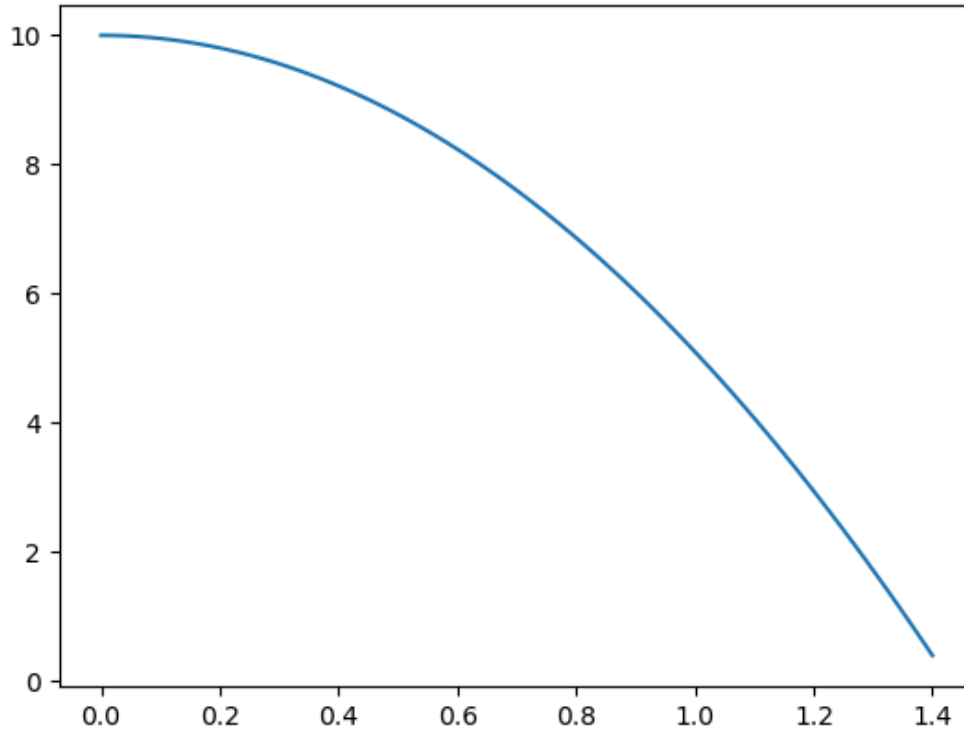
for i in tpoints:
    xpoints.append(r[0])
    vpoints.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6

fig0, ax0 = plt.subplots()
```

```
ax0.plot(tpoints, xpoints)
# ax0.plot(tpoints, vpoints)
```

Out[22]: [<matplotlib.lines.Line2D at 0x7aeb29c0af90>]

Figure



```
In [23]: # define the differential equation
def f(r, t): # don't forget r is a vector!
    x = r[0]
    v = r[1]
    g = 9.8 # N/kg
    m = 1 # kg
    c = 1.5 # N kg s / m ?
    fx = v # this is the dx/dt equation!
    fv = -g - c/m*v # this is the dy/dt equation!
    return(np.array([fx, fv]))

# define the boundary condition
a = 0.0 # starting time
b = 1.414 # ending time
N = 100
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([10.0, 0.0]) # initial condition

tpointsD = np.arange(a, b, dt)
```

```

xpointsD = []
vpointsD = []

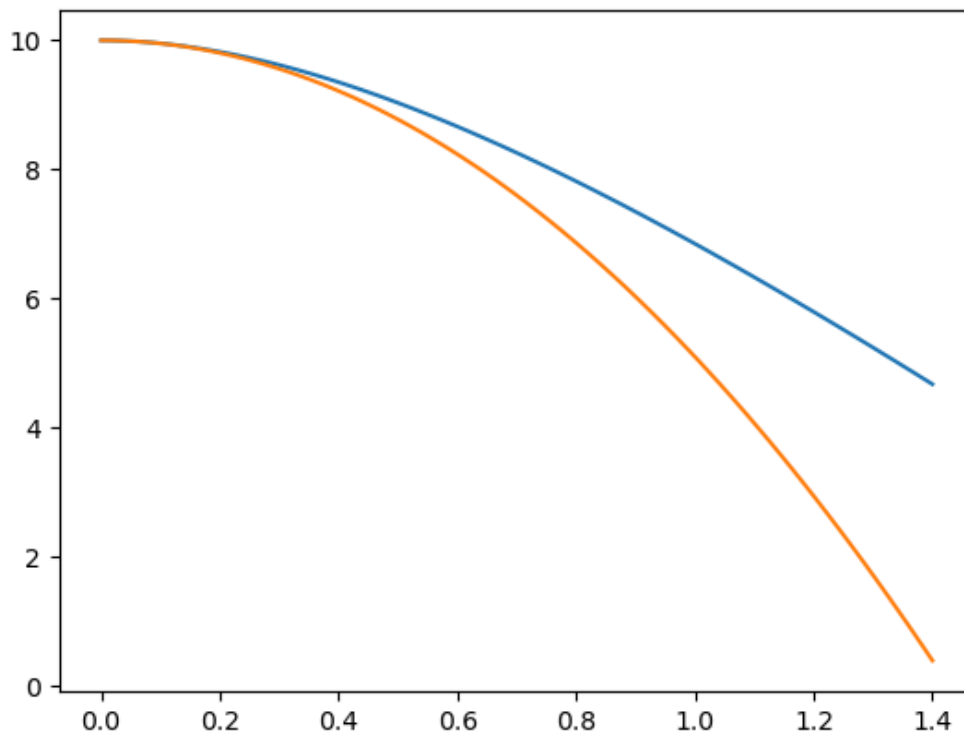
for i in tpoints:
    xpointsD.append(r[0])
    vpointsD.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6

fig0, ax0 = plt.subplots()
ax0.plot(tpoints, xpointsD) # case with drag
ax0.plot(tpoints, xpoints) # case without drag

```

Out[23]: [

Figure



I fixed the code above the next day but I wanted to post it here for next time.

```

In [24]: # define the differential equation
def f(r_, t_): # don't forget r is a vector!
    x = r_[0]
    v = r_[1]
    g = 9.8 # N/kg
    m = 1 # kg
    c = 1.5 # what are these units?

```

```

    fx = v # this is the dx/dt equation!
    if v>=0:
        fv = -g - c/m*v**2 # this is the dv/dt equation!
    else:
        fv = -g + c/m*v**2
    return(np.array([fx, fv]))

# define the boundary condition
a = 0.0
b = 1.414
N = 100
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([10.0, 0.0]) # initial condition

tpointsD2 = np.arange(a, b, dt)

xpointsD2 = []
vpointsD2 = []

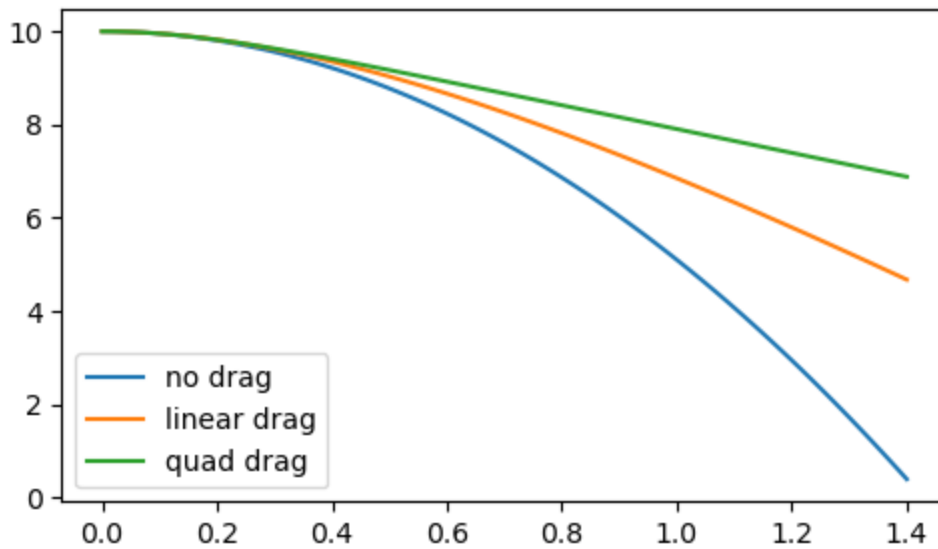
for i in tpointsD2:
    xpointsD2.append(r[0])
    vpointsD2.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6

fig0, ax0 = plt.subplots(figsize=(5,3))
ax0.plot(tpoints, xpoints, label = 'no drag')
fig0.tight_layout()
ax0.plot(tpointsD, xpointsD, label='linear drag')
ax0.plot(tpointsD2, xpointsD2, label = 'quad drag')
ax0.legend()

```

Out[24]: <matplotlib.legend.Legend at 0x7aeb29aa7710>

Figure



```
In [25]: # define the differential equation
def f(r_, t_): # don't forget r is a vector!
    x = r_[0]
    v = r_[1]
    g = 9.8 # N/kg
    m = 1 # kg
    c = 1.5 # what are these units?
    fx = v # this is the dx/dt equation!
    fv = -g - c/m*v*np.abs(v) # this is the dv/dt equation!
    return(np.array([fx, fv]))

# define the boundary condition
a = 0.0
b = 1.414
N = 100
dt = (b-a)/N

# Runge-Kutta 4th order

r = np.array([10.0, 0.0]) # initial condition

tpointsD3 = np.arange(a, b, dt)

xpointsD3 = []
vpointsD3 = []

for i in tpointsD3:
    xpointsD3.append(r[0])
    vpointsD3.append(r[1])
    k1 = dt*f(r, i)
    k2 = dt*f(r+1/2*k1, i+1/2*dt)
    k3 = dt*f(r+1/2*k2, i+1/2*dt)
    k4 = dt*f(r+k3, i+dt)
    r = r + (k1+2*k2+2*k3+k4)/6
```

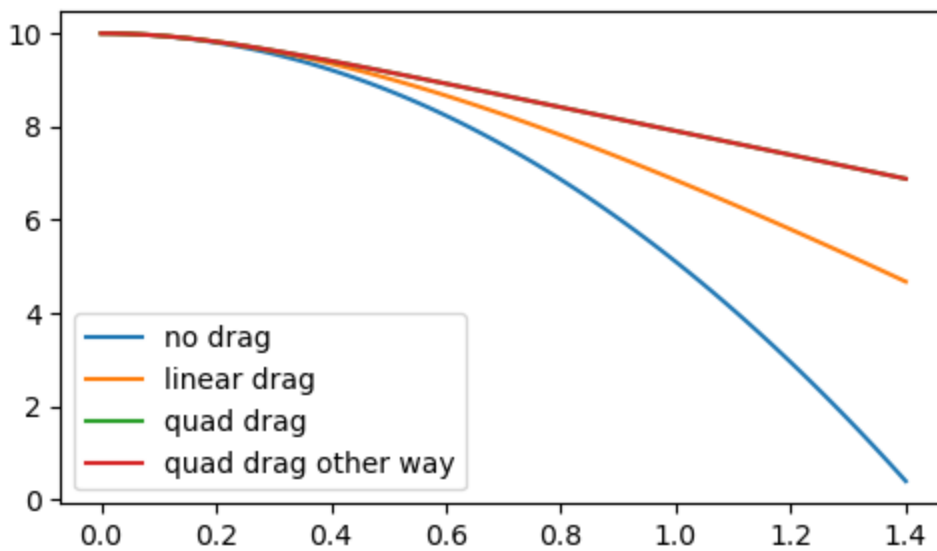
```
fig0, ax0 = plt.subplots(figsize=(5,3))
ax0.plot(tpoints, xpoints, label = 'no drag')
fig0.tight_layout()
ax0.plot(tpointsD, xpointsD, label='linear drag')
ax0.plot(tpointsD2, xpointsD2, label = 'quad drag')
ax0.plot(tpointsD3, xpointsD3, label = 'quad drag other way')
ax0.legend()
```

/tmp/ipykernel_274107/980563573.py:36: RuntimeWarning: More than 20 figures have been opened. Figures created through the pyplot interface (`matplotlib.pyplot.figure`) are retained until explicitly closed and may consume too much memory. (To control this warning, see the rcParam `figure.max\_open\_warning`). Consider using `matplotlib.pyplot.close()`.

```
fig0, ax0 = plt.subplots(figsize=(5,3))
```

Out[25]: <matplotlib.legend.Legend at 0x7aeb29adc260>

Figure



In []:

In []:

In []: